

NeurIPS 2023: How Robust Can a Classifier Possibly Be? The Optimal 0-1 Loss, Now for Multiple Classes

Researchers from Princeton, the University of Chicago, and Penn State compute the best achievable loss against a test-time attacker on any discrete dataset, and measure how far today's robust models still are from it.

Adversarial-robustness research has spent years running an arms race. Someone proposes a stronger defense; someone else designs a stronger attack that breaks it; the cycle repeats. What the loop never supplies is a reference point. How robust can a classifier actually be? Without knowing the theoretical ceiling, there is no way to tell whether a robust model is already near the limit or leaving most of its potential on the table.

For binary classification, prior work established that reference point: given a data distribution and an attacker's budget, the best 0-1 loss any classifier can achieve against a test-time attacker can be characterized exactly. But most real classification problems have many classes, and going from two classes to many is not a routine extension. With three or more classes, examples can interact in higher-order ways that a pairwise, binary analysis simply cannot see, and those interactions can change what optimal robustness even looks like.

This NeurIPS 2023 paper fills that gap. It derives an information-theoretic lower bound on the optimal 0-1 loss for multi-class classification against an L2-constrained test-time attacker on any discrete dataset, and turns it into a tool you can actually compute. The core move is to encode the data and the attacker's constraints as a conflict hypergraph, then solve a linear program whose optimum is the lowest loss any classifier can attain.

Paper and code

Paper: Characterizing the Optimal 0-1 Loss for Multi-class Classification with a Test-time Attacker (NeurIPS 2023), <https://arxiv.org/abs/2302.10722>

Code: https://github.com/inspire-group/multiclass_robust_lb

Why the ceiling is worth computing

The most informative way to judge a robust model is not to compare it against other defenses, but against the best classifier that could possibly exist. If the optimal classifier can only reach 0.4 loss at some attack strength, then a model at 0.45 is close to the limit. But if the optimal loss is 0 and your model plateaus at 0.6, the bottleneck is not the data, it is the training method, and a lot of robustness is going unused.

This bound has two properties that make it useful. It is tied to a specific dataset and threat model, not a generic worst case, and it measures what any classifier can achieve rather than what one particular architecture happens to reach. That lets it separate two questions the arms

race usually conflates: is your method not good enough, or is this problem simply hard? That distinction is exactly what a reference point is supposed to provide.

The core tool: a conflict hypergraph and a linear program

The paper recasts the computation of optimal robust loss as an optimization problem on a graph. The intuition: treat every data point as a vertex. When a group of points from different classes have adversarial neighborhoods that overlap, meaning the attacker could push any of them toward the same confusing input, that group forms a hyperedge. Binary classification only needs pairwise conflicts (ordinary edges); multi-class has to account for three, four, or more points colliding at once, and that is where the higher-order interactions come from.

The figure below shows, on the smallest possible example, why higher-order conflicts matter. Three points come from three different classes, and their neighborhoods overlap pairwise. The difference is whether all three neighborhoods share a common intersection.

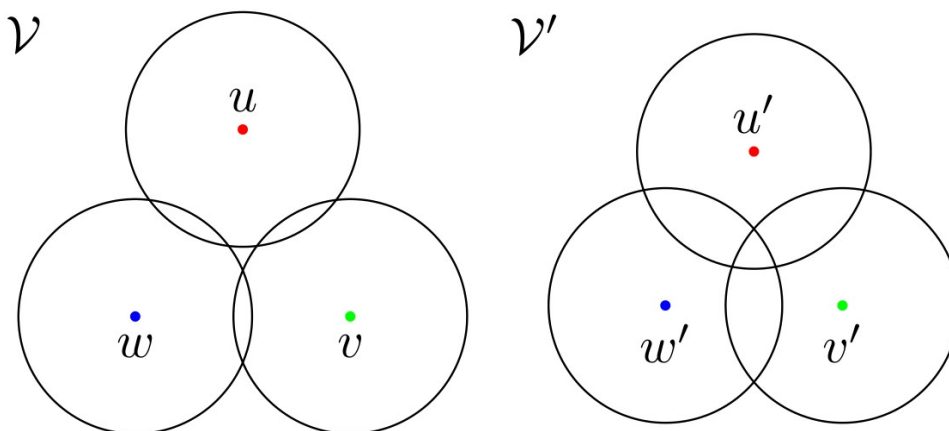


Figure 1. Two conflict structures for three examples, each from a different class. On the left (V), the three neighborhoods overlap pairwise but share no common point, so $\{u, v, w\}$ is not a hyperedge and only the pairwise edges apply. On the right (V'), the three neighborhoods share a common intersection, so every subset of $\{u', v', w'\}$ is a hyperedge. This size-three hyperedge changes the optimal loss only when the three examples' class probabilities are close to balanced (all at most $1/2$), which is why higher-order structure can often be ignored.

Given this hypergraph, the optimal 0-1 loss becomes a linear program: maximize the probability mass of correctly classified points, subject to the constraints encoded by the hypergraph's incidence matrix. Its dual is a fractional cover of the vertices by hyperedges, and that cover corresponds directly to the optimal attacker's strategy. In other words, the same linear program yields, on the primal side, how well the best classifier can do, and on the dual side, how the strongest attacker should play.

The catch is that computing the exact loss requires enumerating hyperedges of every order, which blows up quickly. The paper therefore develops three cheaper bounds that bracket the true optimum in a computable range. Truncating the hypergraph to a maximum order m gives a family of lower bounds $L^*(m)$ (tighter as m grows); aggregating the binary one-versus-one optimal losses via maximum-weight coupling gives a cheaper but looser lower bound $L^{\text{co}}(2)$; and a generalized Caro-Wei approximation of the independent-set number gives an upper bound LCW . Together the lower and upper bounds pin the optimal loss to a narrow interval.

How big is the gap? The 3-class experiments

The authors compute these bounds on two standard vision benchmarks, MNIST and CIFAR-10, under an L2-constrained attacker over a sweep of budgets ϵ . The 3-class setting uses 1000 samples per class (digits 1, 4, and 7 for MNIST; the plane, bird, and ship classes for CIFAR-10). The reference defense is TRADES adversarial training (AT), a three-layer CNN on MNIST and a WRN-28-10 on CIFAR-10, evaluated with the strong APGD-CE attack from AutoAttack.

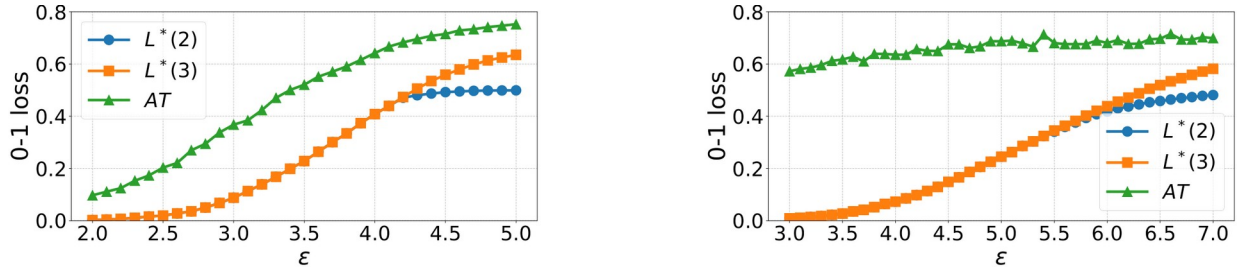


Figure 2. 0-1 loss versus attack strength ϵ on the 3-class problems (left: MNIST digits 1/4/7; right: CIFAR-10 plane/bird/ship). $L^*(2)$ and $L^*(3)$ are the optimal losses (lower bounds) from truncating to degree-2 and degree-3 hyperedges; AT is the measured loss of an adversarially trained model under the strong APGD attack. The adversarial-training curve (green) sits far above the optimal-loss curves throughout: on CIFAR-10, TRADES is stuck near 0.6 loss at a budget where the optimal loss is still close to 0.

This figure delivers the paper's most direct result: there is a large, previously unquantified gap between adversarially trained models and the theoretical optimum, and it is much wider than what was observed in the binary case. On 3-class CIFAR-10, at a budget where the optimal achievable loss is essentially 0, TRADES cannot do better than about 0.6. The headroom left by current robust-training methods is enormous, and it lies in the method, not the data.

Scaling to 10 classes, and a counterintuitive finding

The authors then compute the bounds on the full 10-class problem. The obvious worry is that the low-order truncated lower bounds might be too loose, making the gap above an artifact of slack bounds rather than a real phenomenon. The next figure addresses exactly that.

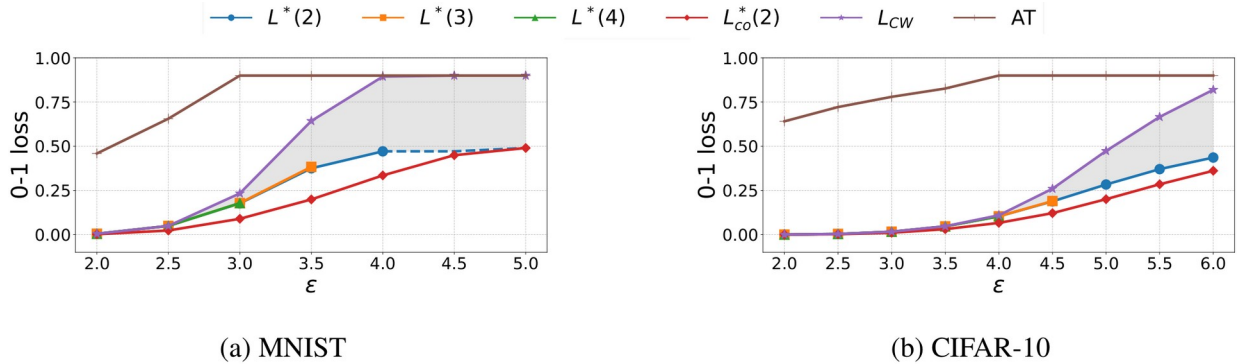


Figure 3. Upper and lower bounds on the optimal loss for the full 10-class problem (left: MNIST; right: CIFAR-10). $L^*(2)$, $L^*(3)$, and $L^*(4)$ are truncated lower bounds of increasing degree, $L^*_{co}(2)$ is the looser aggregated binary bound, and L_{CW} is the Caro-Wei upper bound; the grey band is where the true optimal loss $L^*(10)$ is

expected to fall. At the budgets used in practice the lower and upper bounds sandwich the optimum tightly, showing the adversarial-training gap is not an artifact of loose bounds.

Over the budget range that matters in practice, the truncated lower bounds and the Caro-Wei upper bound nearly coincide, locking the true optimal loss inside a narrow grey band. So the large gap seen in the 3-class case still holds at 10 classes, and it is not an artifact of loose bounds.

More surprising is a finding about how much the higher-order hyperedges actually matter. In principle, multi-class analysis needs to account for degree-3, degree-4, and higher conflicts. But at small attack budgets the lower bound computed from edges alone, $L^*(2)$, is nearly identical to $L^*(3)$ and $L^*(4)$, which add degree-3 and degree-4 hyperedges; the curves only separate once the loss exceeds about 0.4. For CIFAR-10 at $\epsilon=3$, the graph contains roughly 3 million degree-3 and 10 million degree-4 hyperedges, yet they leave the computed bound unchanged. In the low-budget regime that practitioners care about, the edge-only bound is both cheap and accurate.

Putting the gap on the table

Swapping in the best current certifiably robust models does not improve the picture. The table below places their measured 0-1 loss next to the optimal lower bound for the same setting, and the gap is stark.

Table 1. Measured 0-1 loss of the best current certifiably robust models vs. the optimal lower bound (lower is better). Even models designed for robustness with certified guarantees sit far from the theoretical ceiling; in every setting listed, the optimal lower bound is 0.

Setting	Budget ϵ	Best robust model 0-1 loss	Optimal lower bound
MNIST (certified)	1.52	0.27	0
MNIST (certified)	2.0	0.44	0
CIFAR-10 (certified)	1.0	0.6	0
CIFAR-10 (certified)	2.0	0.8	0

Takeaway and boundaries

The paper generalizes the conflict-graph framework to multi-class, characterizing the optimal 0-1 loss against a test-time attacker as a linear program on a conflict hypergraph, with efficient bounds that pin the optimum down in the low-budget regime. With that tool, it quantifies for the first time the gap between adversarially trained multi-class models and the theoretical optimum, and shows it is far worse than in the binary case.

For practitioners, the payoff is a fast diagnostic: compute the optimal-loss interval for your dataset and threat model, and see how far your model is from the ceiling, which tells you whether to keep refining the training method or accept that the problem itself is hard. Worth stating plainly: these results assume discrete data and an L2-constrained test-time attacker, and the numbers depend on the dataset, class choice, and sample size. The lower bound guarantees such a classifier exists; it does not tell you how to train one. But it reframes the goal: rather than iterating attacks and defenses indefinitely, first check how much robustness is still on the table.